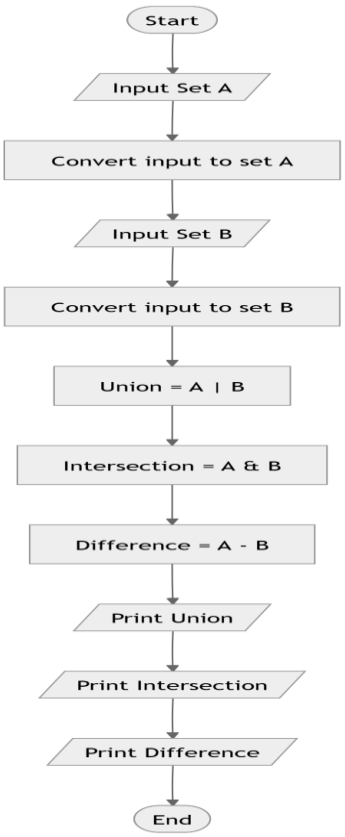
EXPERIMENT 4.1.1

Flowchart



Algorithm:-

Step:-1 Start the program.

Step:-2 Input Set A as space-separated integers from the user.

Step:-3 Convert the input values into a set and store it in seta.

Step:-4 Input Set B as space-separated integers from the user.

Step:-5 Convert the input values into a set and store it in set b.

Step:-6 Compute the union of the two sets using set a | set b.

Step:-7 Compute the intersection of the two sets using set a & set b.

Step:-8 Compute the difference of the two sets using set a – set b.

Step:-9 Print the union of the sets.

Step:-10 Print the intersection of the sets.

Step:-11 Print the difference of the sets.

Step:-12 End the program.

CODETANTRA Home

anushree.kohale.batch2025@sitnagpur.siu.edu.in Support Logout

4.1.1. Set Operations

08:46

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

Explorer

setoperat...

Submit

```python
seta=set(map(int,input("Set A: ").split()))
setb=set(map(int,input("Set B: ").split()))
u = seta | setb
i = seta & setb
d = seta - setb
print("Union:", u)
print("Intersection:", i)
print("Difference:", d)
```

Debugger

Average time	Maximum time
0.009 s	0.012 s
9.50 ms	12.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output	Actual output
Set A: 0 2 4 5 8	Set A: 0 2 4 5 8
Set B: 1 2 3 4 5	Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}	Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}	Intersection: {2, 4, 5}
Difference: {0, 8}	Difference: {0, 8}

Terminal Test cases

Prev Reset Submit Next